Solar Proa: Power Management System

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FYP-IP-01

Problem

As electric vehicle adoption around the world increases, Project Solar Proa aims to develop a carbon-neutral marine vessel. Solar Proa II, the second iteration of the boat at 10 m long, aims to facilitate the development of a small-scale vessel with possible use in tourist attraction and as a novel mode of island transportation.

A robust and reliable power management and monitoring system will be required for the boat to function on the sea for an extended period of time.

Objective

Develops and implements a power management system that captures voltage and current, and other on-board sensors (GPS, Mast Strain, IMU, etc) in real time to be displayed to the crew. The project focuses on developing a prototype to evaluate several state of charge (SoC) estimation methods.



Proa I testing in Changi water

SoC Estimation Methods

4 methods are evaluated: Coulomb Counting, OCV Mapping, Extended Kalman Filter (EKF), and EKF with Kirchhoff's Current Law (KCL) Correction (proposed method).

Prototypes

<u>ESP32 Sensor Network Architecture</u>

- Wireless ESP-NOW network of sensors throughout the vessel

<u>Raspberry Pi Server</u>

- On-board server running offline to collect sensor data and run estimator algorithm
- Opens in any crew member's device connected to the Pi

<u>Custom PCB</u>

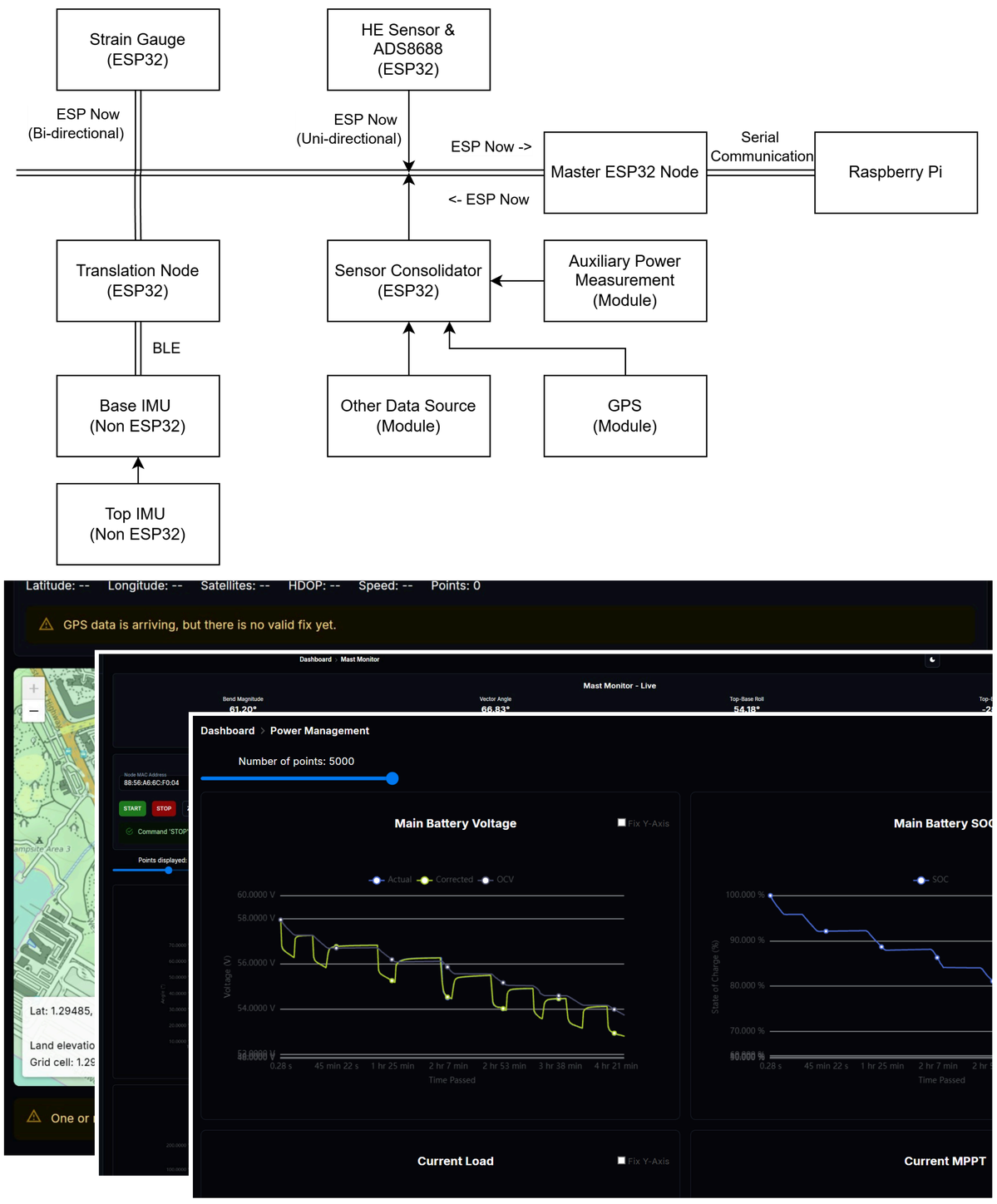
- ESP32 + ADS8688 board to collect voltage and current data

<u>Electrical Component (with PySpice simulation & validation)</u>

- 8x 495W Solar Panel;
- 1x 4,000W, 48V Electrical Motor;
- 2x 50Ah, 25.6V LiFePO4 Battery;
- 2x 100Ah, 25.9V LiNMC Battery;

<u>Electrical Panel</u>

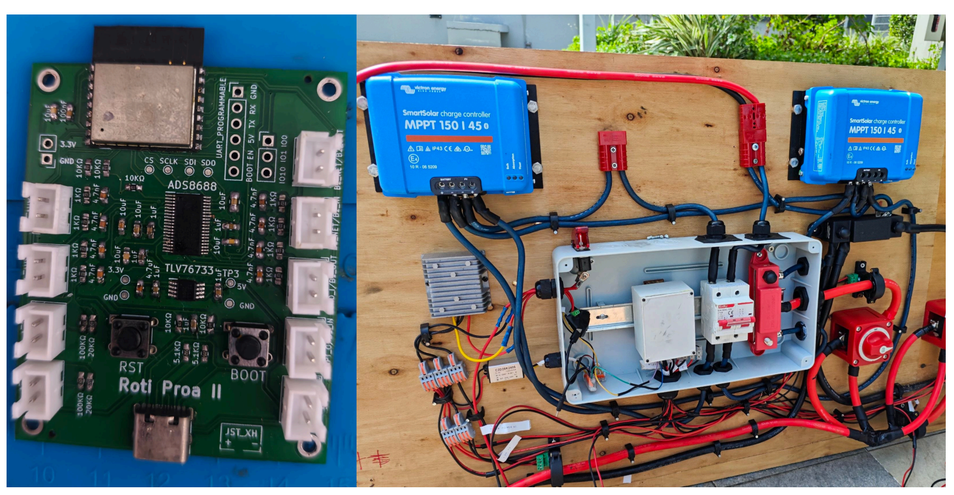
- Mounted electrical panel containing all the essential electrical component, safety, and sensors to run the vessel



Results

EKF significantly reduced sensitivity to initial SoC, achieving a final spread of just 0.008 pp across different initialisation conditions. However, KCL correction had minimal impact on SoC estimates. Moreover, it shares the sensor residual across other sensors, causing offset to incorrectly jump from 0.49 A to 2.62 A. Instead KCL can be used to flag potential sensor error.

Note: These results demonstrate the prototype and EFK's ability to deliver consistent SoC estimation under operational conditions, and not the absolute SoC accuracy.



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